Copy of □ Playwright + AI Mastery - Notes - 2x **Javascript Basics **

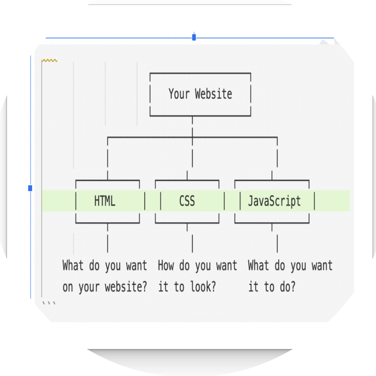
- What is Javascript?** **
- How JavaScript Engine Works
- How does a V8 engine Works?
- Setup IDE Antigravity / VS code
- AI Coding - OpenCode. / Github Copilot Setup.
- Github Account

**What is Javascript? **

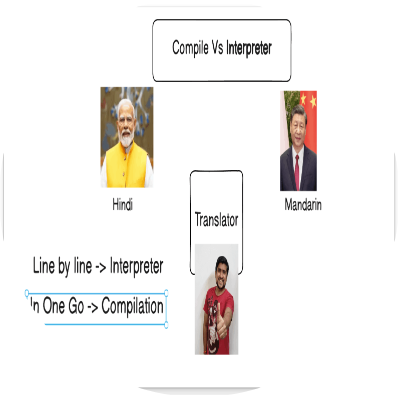
- A website is built with three core technologies
  - **HTML** — Structure / Content
  - **CSS** — Styling / Appearance
  - **JavaScript** — Behavior / Interactivity
- Where it runs: ***Originally designed to run in web browsers,*** JavaScript now also runs on servers (via Node.js), mobile apps, desktop applications, and even IoT devices.
- Core features: **It supports object-oriented, functional, and event-driven programming styles**. It's dynamically typed, meaning you don't need to declare variable types explicitly.
- The** DOM:** **JavaScript can manipulate the Document Object Model (DOM)**, allowing it to dynamically change a webpage's content, structure, and styling in real time without reloading.
- Asynchronous programming: It supports asynchronous operations through callbacks, Promises, and `async/await`, making it efficient for tasks like API calls and file handling.
- Ecosystem: **It has a massive ecosystem of libraries **and frameworks such as React, Angular, Vue.js (frontend) and Express, Next.js (backend), supported by the npm package manager.
- Universal adoption: JavaScript is the most widely used programming language in the world, supported by all modern browsers and backed by a huge developer community.

int a = 10; - data type

In JS -> let a = 10;



Is JavaScript Compiled or Interpreted?



	Compiled Languages	
	Translated to machine code before execution	Executed line by lin
	Generally faster	Generally slower
ortability	Platform-specific	More portable
Development Cycle	Longer (compile-link-execute)	Shorter (immediate execution)
Error Detection	At compile time	At runtime
Memory Usage	More efficient	Higher memory usage
ples	C, C++, Rust, Go	Python, JavaScript, Ruby
	Less flexible	More flexible
	mpiler optimizes code	Limited

JavaScript: Compile-Time & Runtime?

- JavaScript is Primarily a Runtime (Interpreted) Language Traditionally, JavaScript code is executed line by line at runtime by the browser's JS engine.
- Simple Terms - JavaScript is a **runtime language** that uses **JIT compilation internally** for performance.

> *JavaScript is an interpreted language with JIT compilation at runtime.*

Just in time

Browser	JS Engine	Developer
Google Chrome	V8	Google
Microsoft Edge	V8	Google (Chromium)
Mozilla Firefox	SpiderMonkey	Mozilla
Apple Safari	JavaScriptCore (Nitro)	Apple
Opera	V8	Google (Chromium)
Brave	V8	Google (Chromium)
Samsung Internet	V8	Google (Chromium)
Node.js (runtime)	V8	Google
Deno (runtime)	V8	Google
Bun (runtime)	JavaScriptCore	Oven Technologies
Internet Explorer (legacy)	Chakra	Microsoft

Computer - Machine Code

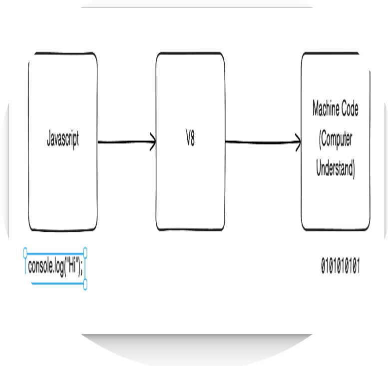
Write Javascript -> How it will run on the Browser?

> *JS Engine*

V8 Engine (Presnet Browsers)

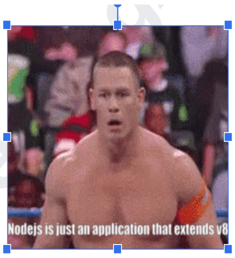
[v8.dev/ (https://v8.dev/)](https://v8.dev/)

V8 is Google’s open source high-performance JavaScript and WebAssembly engine, written in C++.



 What is [Node.js (http://node.js/)](http://node.js/)?

- Node.js is an open-source, cross-platform JavaScript runtime environment that allows you to run JavaScript code outside of a web browser primarily on servers.
- It was created by Ryan Dahl in 2009 and is built on Google Chrome's V8 JavaScript engine.
- Node.js® is a free, open-source, cross-platform JavaScript runtime environment that lets developers create servers, web apps, command line tools and scripts.



Features

Feature	Description
Server-Side JS	Allows JavaScript to run on servers, not just browsers
V8 Engine	Uses Google's fast V8 engine for high performance
Non-Blocking I/O	Handles multiple requests asynchronously without waiting
Single-Threaded	Uses a single thread with an event loop for concurrency
Event-Driven	Responds to events (requests, file reads) efficiently
NPM (Node Package Manager)	World's largest ecosystem of open-source libraries
Platform	Runs on Windows, macOS, and Linux

Common Use Cases

- Web Servers & APIs (REST, GraphQL)
- Real-Time Apps (Chat apps, live notifications)
- Microservices Architecture
- Command-Line Tools (CLI apps)
- Streaming Applications (Video/audio)
- IoT (Internet of Things)
- Mobile Apps

Popular Frameworks Built on Node.js

Framework	Purpose
Express.js	Minimal web server framework
Next.js	Full-stack React framework
NestJS	Enterprise-grade backend apps
Fastify	High-performance web framework
Socket.io	Real-time communication

JavaScript environment is very Big, JavaScript has the most libraries, utilities, and frameworks available compared to any other programming language.

□** NODE js in simple terms**

Node.js = JavaScript + V8 Engine + Server-Side Capabilities, enabling full-stack development using a single language.

JS (Javascript)-> TS(Typescript) -> Playwright (TS)

> *Javascript has nothing to do with JAVA*

How do I write my JavaScript code?

1. NotePad
2. IDE (Integrated Development Environment)
    1. Visual Studio Code
    2. Antigravity - Google
    3. Cursor AI
    4. WindSurf
    5. Eclipse JS
    6. Web Storm - JetBrain

Company Laptop

- Visual Studio Code (Open source FREE, allowd in the 99.9% Copanies)

Personal

- Visual Studio Code
- Antigravity Google

Very Rich (Purchase)

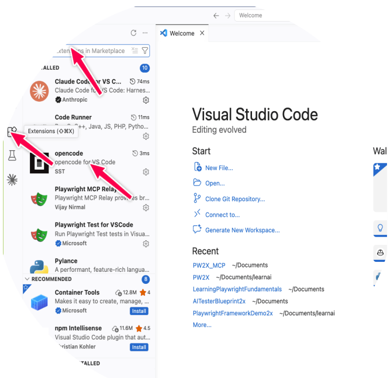
- Cursor AI, Windsurf.
- Web Storm
- Codex
- Claude Code

□ Install Visual Studio Code

[code.visualstudio.com/download (https://code.visualstudio.com/download)](https://code.visualstudio.com/download)

Option 1 : Visual Studio Code + Github Coplit (Company Has given) - Please use this

Option 2 : Visual Studio Code + OpenCode (Go Subscrition first Month 484 Rs) per month - Kimi k2.6 -~ Claude Opus 4.5



☐ **Install Google Antigravity **

**Disclaimer - **Try to avoid to install in Company Laptop

antigravity.google/ (https://antigravity.google/)

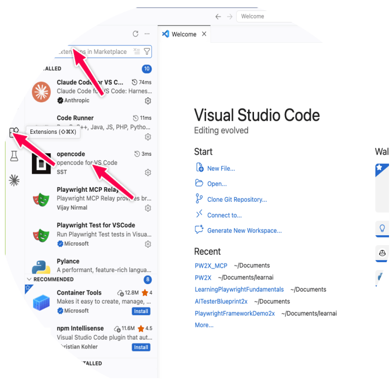
How to set up the open code in Visual Studio Code?

- just install the Visual Studio Code, install the extension, and just log into your open code subscription opencode.ai/go (https://opencode.ai/go)

You can buy the open code subscription, go for Rs 484 per month for the first month. Please rotate the email ID when it is required after the second month, which is there. You can use Deep Seek v4 Pro or Kimi K2.6 version, whichever both of them I use.

Install this extension into the Visual Studio Code, and you just have to log in to open code, and you are good to go.

marketplace.visualstudio.com/items?itemName=sst-dev.opencode (https://marketplace.visualstudio.com/items?itemName=sst-dev.opencode)



Install Node.js

Windows:

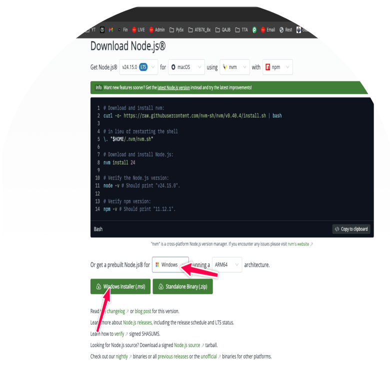
1. Download the Windows installer from the Node.js website (https://nodejs.org/).

2. Run the installer.

nodejs.org/en/download (https://nodejs.org/en/download)

3. Follow the prompts in the installer (Accept the license agreement, click the NEXT button a bunch of times and finally install).

4. Restart your computer to ensure changes take effect.



MacOS:

1. Download the macOS installer from the Node.js website (https://nodejs.org/).

2. Open the `.pkg` file you downloaded and follow the instructions.

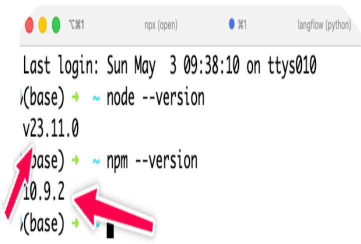
3. You may need to enter your admin password.

4. To verify installation, open a terminal and type `node -v`.

node --version
npm --version

Windows -> ** cmd**

Mac /LINUX -> **Terminal**



Task for the Today 4th May 2026

Task 1 - Install Visual Studio Code / Antigravity

Task 2 - Install the Node JS (verify)

Task 3 - Create the github.com/signup

Download Google Antogravity - antigravity.google/ (https://antigravity.google/)

Please install Google Anti-Gravity on your personal laptop if you don't want to use Visual Studio Code. Both Visual Studio Code and Anti-Gravity are the same

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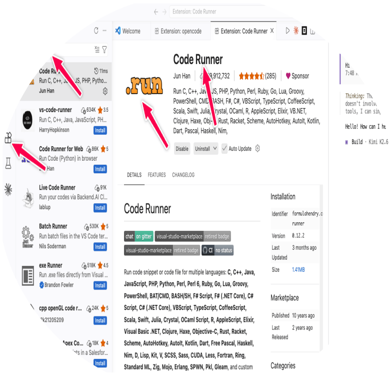
To run the JavaScript code

in the Visual Studio Code, we need the Code Runner extension

marketplace.visualstudio.com/items?itemName=formulahendry.code-runner (https://marketplace.visualstudio.com/items?itemName=formulahendry.code-runner)



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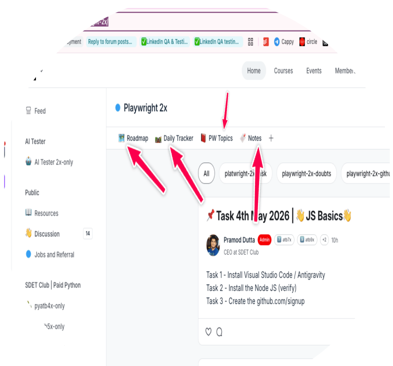
1. Everyone should have access to the [sdetclub.com (https://sdetclub.com/)](https://sdetclub.com/) -> Fill this form - to get access to the SDETCLUB - [docs.google.com/forms/d/e/1FAIpQLSdr8F8m7jmIl6pdqezE0LZ8kphE2-ZpUTjBc9YxSBbsUjONaA/viewform (https://docs.google.com/forms/d/e/1FAIpQLSdr8F8m7jmIl6pdqezE0LZ8kphE2-ZpUTjBc9YxSBbsUjONaA/viewform)](https://docs.google.com/forms/d/e/1FAIpQLSdr8F8m7jmIl6pdqezE0LZ8kphE2-ZpUTjBc9YxSBbsUjONaA/viewform)
    1. SDET Club we are using for the
        1. Doubts
        2. TASK
        3. LIVE TEST
        4. Notes

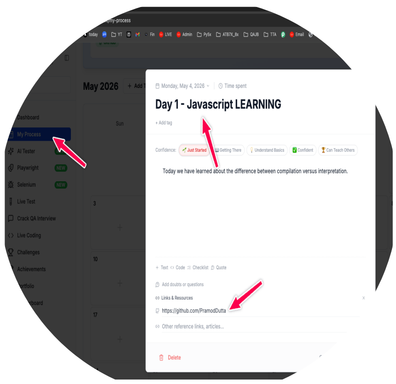
This is your own Private Group - [www.sdetclub.com/c/playwright-2x/ (https://www.sdetclub.com/c/playwright-2x/)](https://www.sdetclub.com/c/playwright-2x/)

[docs.google.com/forms/d/e/1FAIpQLSdr8F8m7jmIl6pdqezE0LZ8kphE2-ZpUTjBc9YxSBbsUjONaA/viewform (https://docs.google.com/forms/d/e/1FAIpQLSdr8F8m7jmIl6pdqezE0LZ8kphE2-ZpUTjBc9YxSBbsUjONaA/viewform)](https://docs.google.com/forms/d/e/1FAIpQLSdr8F8m7jmIl6pdqezE0LZ8kphE2-ZpUTjBc9YxSBbsUjONaA/viewform)



How to mark attendance?

1. create a free account on this - [app.thetestingacademy.com/student/my-process (https://app.thetestingacademy.com/student/my-process)](https://app.thetestingacademy.com/student/my-process)
2. on today's date, you need to mention your title and topic, whatever you have learned, as well as if you have created your GitHub repository. Also mention that and click on Save Entry



□ How JavaScript Engine Works

A JavaScript engine is a program that **reads, interprets, and executes** JavaScript code.

Here's a complete breakdown of how it works step by step.